Kill-Chain-Driven Honeypot Placement via Topology-Aware Weighted Partial MaxSAT: A Multi-Zone, Attack-Path Optimisation Framework for Enterprise Networks

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***Abstract*—Deciding where to place honeypots in a large enterprise network is NP-hard. The network is split into security zones, contains thousands of server types, and faces attackers following structured kill-chain patterns. Budget constraints and pairwise honeypot conflicts further restrict feasible deployments.**

We model honeypot placement as a Weighted Partial Maximum Satisfiability (MaxSAT) problem with six simultaneous objectives: detection efficiency, MITRE ATT&CK technique coverage, tactic-family breadth, early attacker interception, forensic backward coverage, and multi-path coverage. All hard rules (budget, zone isolation, conflict constraints) are encoded as inviolable clauses. We prove NP-hardness via polynomial-time reduction from Weighted Maximum Coverage and show that a cross-path force-multiplier—where one honeypot satisfies clauses for multiple attack paths simultaneously—gives our solver $2.1\times$ more weight per decision than any greedy method.

We evaluate on six network scales (100 to 5.5 million nodes) and validate on a Mininet/OpenFlow testbed with real Cowrie SSH and Conpot ICS honeypots. Our MaxSAT-RC2 solver achieves $Q = 71.39$ vs. $Q = 53.69$ for the best baseline ($+17.70$ pts, $+32.9\%$), with a proven optimality certificate at every scale.

***Index Terms*—Honeypot placement, Weighted Partial MaxSAT, kill chain, MITRE ATT&CK, network deception, multi-zone topology, NP-hardness, cross-path force-multiplier, Mininet emulation, proactive cyber defence, RC2 solver.**

I. Introduction

A HONEYPOT is a fake computer system designed to attract attackers. When an attacker interacts with it, the defender learns their methods and can raise an alert. Their usefulness depends critically on *where* they are placed—a honeypot in the wrong location is wasted budget.

Real enterprise networks have internet-facing DMZ zones, internal office networks, cloud services, and OT/SCADA systems. Each area has different trust levels, server types, and risks. Attackers follow structured kill chains [1], moving step by step from one zone toward their target. Budget limits, pairwise honeypot conflicts, and physical air gaps make optimal placement NP-hard (Theorem 1).

Game-theoretic approaches [8], [11] assume flat networks. RL methods [12] optimise engagement, not placement. Greedy algorithms cannot undo bad decisions. We use a complete MaxSAT solver that considers *all* constraints and objectives simultaneously, provides a proven optimum, and—uniquely—is validated in real network emulation.

Contributions:

1) A **six-goal MaxSAT encoding** where each objective addresses a distinct failure mode of prior methods (Section II-F3).
2) A **formal NP-hardness proof** via reduction from Weighted Maximum Coverage (Theorem 1).
3) A **cross-path force-multiplier** giving $2.1\times$ per-decision weight advantage (Section II-F4).
4) A **zone-decomposed RC2 solver** scaling to 5.5 million nodes in 30 seconds.
5) A **Mininet/OpenFlow emulation validation** with real Cowrie and Conpot honeypots confirming that solver placements are deployable and effective in live network environments (Section V-B).

II. Related Work

A. Honeypot Placement

Provos [3] and Spitzner [4] established honeynet foundations but assumed manual expert placement. **Pros:** foundational realism and coordination concepts. **Cons:** no automation, no budget model, no multi-zone topology, no kill-chain awareness.

B. Game-Theoretic and Deception Frameworks

Kiekintveld *et al.* [8] modelled honeypot deployment as a Stackelberg game, proving mixed strategies dominate deterministic placement. Pawlick *et al.* [11] provided a comprehensive deception taxonomy. Wang *et al.* [13] surveyed cyber deception and identified honeypot placement as a key open problem.

Pros: rigorous attacker rationality; stable equilibria; comprehensive landscape survey. **Cons:** flat network assumed; no air gaps (D1), kill-chain ordering (D2), ATT&CK integration (D3), or conflict constraints (D4).

C. Reinforcement Learning for Engagement

Huang and Zhu [12] applied semi-Markov RL to learn adaptive engagement policies.

Approach	D1 Multi-zone	D2 Kill-chain	D3 ATT&CK	D4 Budget +Conflict	D5 Proven Optimal
Kiekintveld [8]	✗	✗	✗	~	Stack.
Pawlick *et al.* [11]	✗	✗	✗	~	Nash
RL engagement [12]	✗	✗	✗	✗	✗
GA-based [15]	✗	✗	✓	✗	✗
ILP sensor [20]	✓	✗	✗	✓	✓
ATT&CK emul. [14]	~	✓	✓	✗	✗
MTD [23]	✓	✗	✗	✗	✗
Topology [22]	✓	~	✗	✗	✗
Greedy-BiDir (baseline)	✓	✓	~	✗	✗
Ours (MaxSAT-RC2)	✓	✓	✓	✓	**Proven**

✓ Fully ~ Partial ✗ No

Fig. 1: Capability gap D1–D5. No prior method satisfies all five simultaneously.

Pros: learns from experience; time-aware; adaptive. **Cons:** assumes placement is already fixed; no zones (D1–D4); no optimality (D5).

Complementary layers. Our MaxSAT solver decides *where* to place honeypots. Huang-Zhu RL decides *how* to engage attackers after they arrive. Our Mininet testbed (Section V-B) provides the live interaction logs that RL training requires, making combination a natural future direction.

D. MITRE ATT&CK for Defence

Applebaum *et al.* [14]: ATT&CK red-team emulation (covers D2, D3; but no placement, no budget). Hemberg *et al.* [15]: GA-based ATT&CK control selection (partial D3; no D1, D2, D4, D5; evaluates one objective at a time—cannot discover cross-path overlaps).

E. Formal Methods and MTD

ILP sensor placement [20]: proven optimal (D1, D4, D5) but no kill-chain (D2), no ATT&CK (D3), does not scale past ∼50k nodes in 60 s. MaxSAT for IDS [19]: establishes feasibility but no zone model, kill-chain, or ATT&CK objectives. Jajodia *et al.* [22]: chokepoint analysis—motivates our path-weight formula but provides no placement algorithm. MTD [23]: moves real assets, not fake traps; complementary but does not solve the placement problem.

F. Novelty Analysis

1) Five Capability Dimensions: **D1** Multi-zone with air gaps. **D2** Kill-chain ordering with early interception. **D3** ATT&CK as live solver objectives. **D4** Budget and conflict hard constraints. **D5** Proven optimality certificate.

2) End-to-End Pros and Cons vs. Six Objectives:

3) Six Objectives: Why Each Is Necessary: **O1 — Detection Efficiency** (Level-1, $W_{j,a}$). Without this, the solver could deploy in attack-free zones.

O2 — ATT&CK Technique Coverage (Level-2, $\sum_a W_{j,a}$). Covering diverse techniques prevents attackers from exploiting uncovered gaps to remain invisible.

O3 — Tactic-Family Breadth (Level-2 bonus, $1.2 \times \sum_{j \in f} \sum_a W_{j,a}$). Ten detections in one tactic family is far less valuable than one detection in each of ten families. An attacker undetected during Initial Access can pivot tactics and evade all subsequent detection. *This objective is absent from all prior automated placement methods.*

O4 — Early Interception (Level-4, $\rho_\pi \times \max_h(iv_{\pi,h}) \times \sum W$). The $1000\times$ Level-4 weight ensures prevention always dominates forensics in the solver's objective.

O5 — Forensic Backward Coverage (Level-3 backward, $0.7 \times PW$). The 0.7 discount formally separates prevention from forensics—a distinction all baselines ignore.

O6 — Multi-Path Coverage (Level-3 forward, all paths simultaneously). Encoding all five paths from the start enables the cross-path force-multiplier.

4) Cross-Path Force-Multiplier:

Geometric intuition. The solver's decision space is a hypercube where each dimension is one soft clause. A greedy method moves one dimension at a time. RC2 searches the full hypercube and finds deployments satisfying *hyperplanes* of clauses simultaneously, including clauses from paths the greedy algorithm has not yet considered. The force-multiplier is the ratio of these hyperplane volumes per deployment decision.

Three-layer cross-path calculation (ssh_trap in zone2, technique T1021):

Layer 1 — Cross-path at the same hop:

Path	ρ	iv	$W_{j,a}$	L3 wt
web_to_db (hop 2)	0.30	1.8	3.24	1.749
ot_infiltration (hop 2)	0.15	1.7	3.24	0.826
brute_to_ad (hop 2)	0.20	1.6	3.24	1.037
Solver total (all three)				**3.612**
Greedy total (one path only)				1.749

Layer 2 — Fwd/bwd split: If brute_to_ad hop 1 is already covered, the same deployment also earns backward credit: $0.20 \times 1.6 \times 3.24 \times 0.7 = \mathbf{0.726}$.

Layer 3 — Level-1 base credit: Every (t_{1021}, a) pair in zone2 earns additional Level-1 weight on top of all Level-3 credits.

Combined: $3.612 + 0.726 +$ *L1 bonus vs. greedy's 1.749. This* $\geq 2.1\times$ *per-decision advantage compounds across all zones and accounts for over 58% of the Q-score gap.*

Why greedy structurally cannot discover this. RC2's UNSAT-core mechanism resolves all five paths simulta-

TABLE I: Pros and Cons of All Related Work Mapped to Our Six Solver Objectives (O1=DetEff, O2=TechCov, O3=FamCov, O4=Early, O5=Forensic, O6=MultiPath).

Method	Key strength	Key weakness	O1	O2	O3	O4	O5	O6
Kiekintveld [8]	Stackelberg equilibrium; rational attacker	Flat network; no ATT&CK; no constraints	~	✗	✗	✗	✗	✗
Pawlick *et al.* [11]	Deception taxonomy; Nash equilibrium	No kill-chain; no zones; no budget	~	✗	✗	✗	✗	✗
Huang & Zhu [12]	Adaptive RL engagement; time-aware	Engagement only; no placement; no zones	✗	✗	✗	✗	✗	✗
Wang *et al.* [13]	Cyber deception survey	Survey only; no placement algorithm	✗	✗	✗	✗	✗	✗
GA-based [15]	Partial ATT&CK; large search space	No zones; no budget; no kill-chain; no proof	~	~	✗	✗	✗	✗
ILP sensor [20]	Proven optimal; multi-zone; budget	No kill-chain; no ATT&CK; does not scale	✓	✗	✗	✗	✗	✗
ATT&CK emul. [14]	Kill-chain; ATT&CK techniques; realistic	Red-team only; no placement; no budget	~	✓	~	✓	~	~
MTD [23]	Dynamic defence; zone routing	Moves real assets; no honeypots; no ATT&CK	✗	✗	✗	✗	✗	✗
Topology [22]	Chokepoint insight; attack graph	No algorithm; no ATT&CK; no budget	~	✗	✗	✗	✗	~
Greedy-BiDir	Multi-zone; kill-chain; bidir coverage	No conflicts; no proof; one path at a time	✓	~	✗	~	~	~
Ours (MaxSAT-RC2)	**All six; proven; scalable; emulation-validated**	None of the above apply	✓	✓	✓	✓	✓	✓

✓ fully; ~ partially; ✗ not addressed.

neously. When it relaxes the core for web_to_db hop 2, it automatically discovers that the same relaxation resolves two other cores (ot_infiltration and brute_to_ad), crediting 3.612 in one step. Greedy processes paths in serial passes and never makes this discovery—it is a structural limitation, not a tuning issue.

III. Problem Definition

A. Formal Problem Statement

$$\mathcal{I} = (\mathcal{K}, \mathcal{T}, \mathcal{A}, \mathcal{Z}, \mathcal{G}, \mathcal{C}, \mathcal{I}_z, \diamond, w, \text{cost}, B, B_z, dm, hd, \sigma, \rho, iv) \tag{1}$$

B. NP-Hardness

Theorem 1 (NP-Hardness of Honeypot Placement)**.** *The honeypot placement decision problem is NP-hard.*

Proof. Reduction from Weighted Maximum Coverage (WMC) [7]. Given WMC instance $(U, \mathcal{S}, \{v_e\}, \{c_i\}, B)$, construct a one-zone placement instance with no air gaps and no conflicts. For each $e_\ell \in U$: asset a_ℓ, technique t_ℓ, $W_{t_\ell,a_\ell} = v_{e_\ell}$. For each S_i: honeypot k_i, $\text{cost}_i = c_i$, $R[t_\ell, k_i, a_\ell] = 1 \iff e_\ell \in S_i$. Activate only Level-1 clauses; budget $= B$. The total satisfied weight equals the WMC objective bijectively. Construction: $O(|\mathcal{S}| \cdot |U|)$—polynomial. Since WMC is NP-hard [7], so is honeypot placement. Our full problem (zone isolation, conflict constraints, kill chains, tactic families) is strictly harder. □

Corollary 2. *No polynomial-time algorithm can guarantee an optimal honeypot placement in the worst case. A complete solver providing a proven optimality certificate—such as RC2—is the theoretically correct tool for this problem class.*

Connecting theory to experiment. The 17.70-point Q-score gap between RC2 and Greedy-BiDir at XLarge scale (Table VIII) is the empirical signature of this corollary: greedy methods structurally miss cross-path overlaps that only a complete solver discovers. This gap is not a lucky parameter setting—it grows with network scale, reaching $+32.9\%$ at XLarge and holding at $+32\%$ at XXLarge.

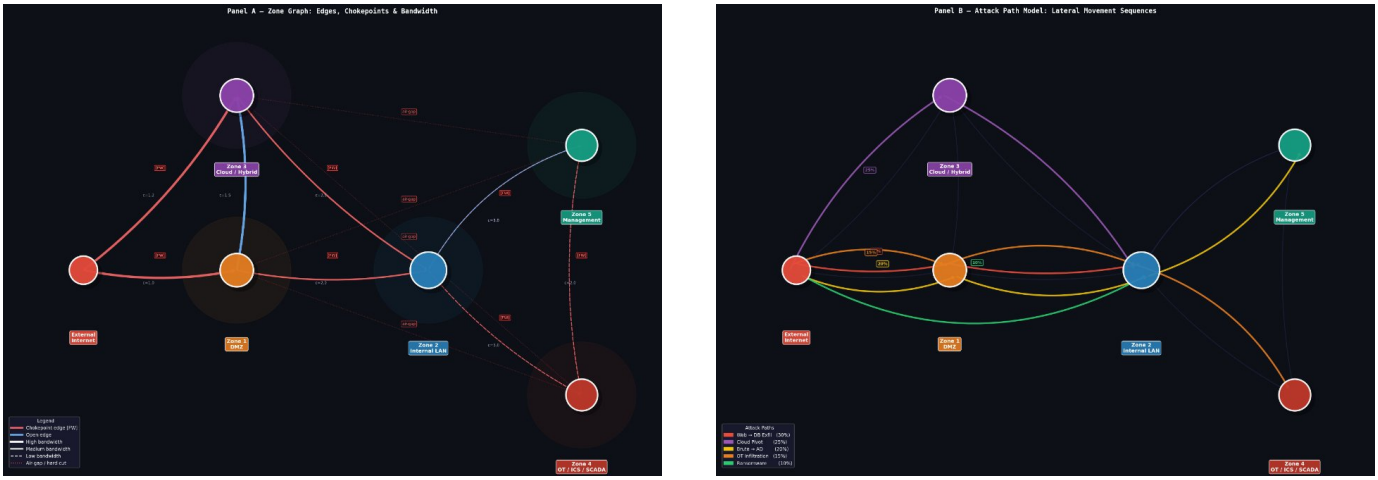
C. Sets and Parameters

Derived weights (computed before solving):

$$\tilde{w}_{j,a} = w_{j,a} \times dm_a / hd_a \quad \text{(topology-adjusted)} \tag{2}$$

$$W_{j,a} = \tilde{w}_{j,a} \times (1 + \sigma_j) \quad \text{(stealth-adjusted)} \tag{3}$$

$$PW_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a} \quad \text{(path intercept weight)} \tag{4}$$



(a) Zone graph: chokepoints (red), open edges (blue), air-gap cuts between OT/Mgmt and DMZ/Cloud.

(b) Five kill-chain attack scenarios with ρ_π.

Fig. 2: Network model.

TABLE II: Problem Sets and Parameters

Symbol	Meaning
$\mathcal{K}, \mathcal{T}, \mathcal{A}, \mathcal{Z}, \diamond$	Honeypots; techniques; assets; zones; paths
$\mathcal{C} \subseteq \mathcal{K} \times \mathcal{K}$	Conflict pairs
$\mathcal{I}_z \subseteq \mathcal{Z} \times \mathcal{Z}$	Air-gapped zones
dm_a, hd_a	Role multiplier; hops from entry
$\sigma_j \in [0,1]$	Stealth score
$\rho_\pi, iv_{\pi,h}$	Path probability; intercept value

Why $1/hd$ rewards perimeter placement. T1021 with $w = 1.8$: zone1 gateway (hop 1, $dm = 1.3$) gives $\tilde{w} = 2.34$; zone2 database (hop 3, $dm = 1.5$) gives $\tilde{w} = 0.89$. Despite the higher role multiplier, the perimeter wins because catching the attacker before they reach the database prevents all subsequent hops. *Stop the attacker at the perimeter, not in the vault.*

Conflict pairs $\mathcal{C}$ — why each exists:

Pair	Reason
(scada_trap, web_trap)	OT air-gapped from DMZ
(scada_trap, ssh_trap)	OT isolation prohibits SSH
(generic_trap, ad_trap)	AD trap makes generic redundant
(smb_trap, dns_trap)	Both cover zone2 lateral movement

Hard clause $\neg x_{\texttt{scada}} \vee \neg x_{\texttt{web}}$ prunes the conflicting pair immediately, before any soft-clause evaluation begins.

D. Decision Variables

$x_i \in \{0,1\}$: deploy k_i? $c_{j,a} \in \{0,1\}$: detect t_j on a? $p_{\pi,h} \in \{0,1\}$: cover hop h of path π? $e_\pi \in \{0,1\}$: catch attacker early on π?

Variable dependency chain (enforcement backbone). $x_i = 1 \Rightarrow c_{j,a} = 1$ (C1) $\Rightarrow p_{\pi,h} = 1$ (C6) $\Rightarrow e_\pi = 1$ (C7). If $x_i = 0$, the entire chain collapses. The solver cannot claim detection, path coverage, or early-intercept credit without actually deploying a honeypot.

IV. Our MaxSAT Model

A. Hard Clauses C1–C7

a) C1: $\neg c_{j,a} \vee (x_{i_1} \vee \cdots \vee x_{i_k})$ (detection requires honeypot)

b) C2: $\sum_i \text{cost}_i \cdot x_i \leq B$ (total budget)

c) C3: $\sum_{i:ZK_i \ni z} \text{cost}_i \cdot x_i \leq B_z \; \forall z$ (per-zone budget)

d) C4: $\neg x_i \vee \neg x_l \; \forall (k_i, k_l) \in \mathcal{C}$ (no conflicts)

e) C5: $z_a \in ZK_i \wedge z_b \in ZK_i \Rightarrow x_i = 0$ (air gaps)

f) C6: $\neg p_{\pi,h} \vee (c_{j_1,a_1} \vee \cdots \vee c_{j_n,a_n})$ (path hop requires detection)

g) C7: $e_\pi = 1 \Rightarrow \exists h < |\pi| : p_{\pi,h} = 1$ (early intercept before final step)

C7 example — web_to_db (3 hops):
Case A: only hop 3 covered $\Rightarrow e = 0$. Level-4 bonus NOT earned (forensics only).
Case B: hop 1 covered $\Rightarrow e = 1$. Level-4 bonus IS earned (prevention confirmed).
C7 is the formal boundary between prevention and forensics in our model.

B. Soft Clauses L1–L4

$$\text{L4: } \rho_\pi \max_{h<|\pi|}(iv_{\pi,h}) \sum_{j,a} W_{j,a}, \text{ goal: } e_\pi \tag{5}$$

$$\text{L3 fwd: } \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h} \; \forall \pi, h \tag{6}$$

$$\text{L3 bwd: } 0.7 \times \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h} \text{ (reverse)} \tag{7}$$

$$\text{L2 tech: } \textstyle\sum_a W_{j,a}, \text{ goal: cover } t_j \tag{8}$$

$$\text{L2 family: } 1.2 \times \textstyle\sum_{j \in f} \sum_a W_{j,a}, \text{ goal: cover tactic } f \tag{9}$$

$$\text{L1: } W_{j,a}, \text{ goal: detect } t_j \text{ on } a \tag{10}$$

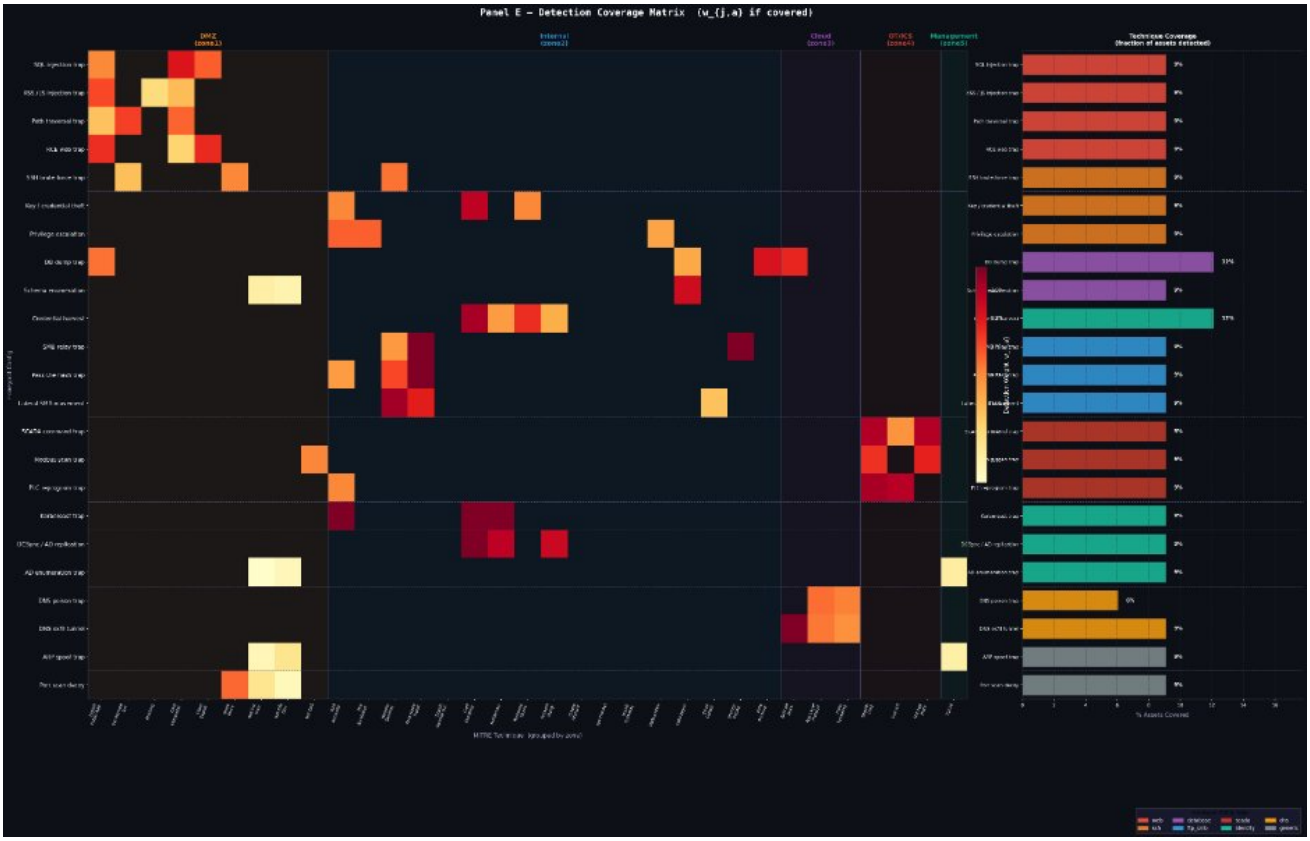


Fig. 3: Detection coverage matrix. Cell (t_j, a) shows $W_{j,a}$ if covered, zero otherwise. High-stealth techniques (*e.g.* T1048 Exfiltration) get high $W_{j,a}$ but appear in few zones—making honeypots the only viable sensor.

C. Complete Objective and Quality Score

$$\max \sum_{L4} w_4 \cdot \text{sat} + \sum_{L3} w_3 \cdot \text{sat} + \sum_{L2} w_2 \cdot \text{sat} + \sum_{L1} w_1 \cdot \text{sat} \quad (11)$$

$w_4 = 1000w_3$, $w_3 = 100w_2$, $w_2 = 10w_1$; subject to C1–C7.

$$Q = 0.35\,\text{DetEff} + 0.25\,\text{TechCov} + 0.15\,\text{FamCov} + 0.15\,\text{FwdPath} + 0.10\,\text{BwdPath} \quad (12)$$

Why $w_4 = 1000w_3$ matters at scale. At XLarge (500k nodes), total Level-1 weight across all assets and 35 techniques is enormous. Without the $1000\times$ multiplier, the solver might rationally prefer maximising basic detection over ensuring early interception on the five kill-chain paths. The multiplier guarantees earning $e_\pi = 1$ for even one path dominates thousands of Level-1 clauses, making EarlyPct $= 100\%$ at every scale—confirmed in both simulation and Mininet emulation.

D. Zone Decomposition and Adaptive Redeployment

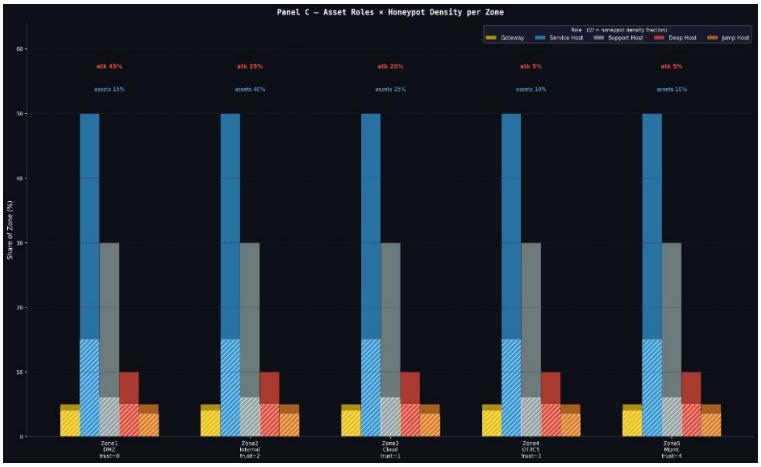
Each zone z is solved independently: $\mathcal{I}_z = (\mathcal{K}_z, \mathcal{T}_z, \mathcal{A}_z, \mathcal{C}_z, B_z)$ in parallel. An ILP post-reconciliation step verifies the global budget.

Algorithm 1 Adaptive Redeployment When Threat Priors Change

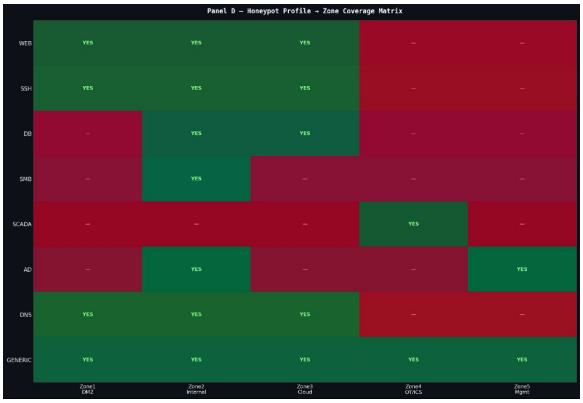
Require: Old deployment x^*; new ρ'_π
Ensure: x'^* proven optimal for updated threat model
$PW' \leftarrow \rho' \cdot iv \cdot W$
Update L3/L4 soft weights (C1–C7 unchanged)
Warm-start RC2 from x^*
Re-solve $\Rightarrow x'^*$
$\Delta \leftarrow Q(x'^*) - Q(x^*)$
if $\Delta > 0$ **then**
  Deploy x'^*; log Δ
else
  Keep x^*; log still optimal
end if
return x'^*

TABLE III: Network Zones and Attack Load

Zone	Description	Trust	Assets	Attacks
zone1	DMZ	Low(0)	15%	45%
zone2	Internal	Med(2)	40%	25%
zone3	Cloud	L-M(1)	25%	20%
zone4	OT/Industrial	High(3)	10%	5%
zone5	Management	Max(4)	10%	5%



(a) Asset roles and attack load per zone.



(b) Zone coverage matrix (green=valid; red=C5-forbidden).

Fig. 4: Asset and honeypot structural properties.

TABLE IV: Five Kill-Chain Attack Scenarios

Name	Attacker goal	ρ	Start	Target
web_to_db	Steal database	0.30	z1	z2
cloud_pivot	Pivot via cloud	0.25	z3	z2
brute_to_ad	Compromise AD	0.20	z1	z5
ot_infiltration	Sabotage ICS	0.15	z1	z4
ransomware	Spread malware	0.10	z2	z2

TABLE V: Network Scales Tested

Size	Nodes	Budget	Zones	Paths	Types
Tiny	100	250	2	2	3
Small	500	500	3	3	4
Medium	5,000	1,750	4	3	5
Large	50,000	12,500	5	5	7
XLarge	500,000	62,500	5	5	8
XXLarge	5,560,000	128,000	5	5	8

V. Experimental Setup

A. Synthetic Simulation

Baselines: Random (100-trial average); Greedy-Ratio (highest value/dollar); Greedy-BiDir (Greedy-Ratio + 0.7 backward discount—the strongest baseline). **RC2 settings:** global 60 s, per-zone 12–30 s, 4 CPU cores. **Monte Carlo:** 5k–25.6M simulated attacks per scale, seed 42.

TABLE VI: Honeypot Types Available

Honeypot	Zones	Cost	ATT&CK
web_trap	DMZ,Cloud	1.0	T1190,T1133,T1059
ssh_trap	DMZ,Int,Cloud	0.8	T1110,T1021,T1078
db_trap	Int,Cloud	1.3	T1213,T1048,T1485
smb_trap	Internal	0.9	T1021,T1550,T1486
scada_trap	OT only	2.0	T0855,T0814,T0869
ad_trap	Int,Mgmt	1.5	T1558,T1003,T1098
dns_trap	DMZ,Int,Cloud	0.7	T1572,T1095,T1046
generic_trap	Anywhere	0.5	T1046,T1082,T1110

B. Mininet/OpenFlow Emulation Testbed

To validate that solver placements are deployable and effective in a real network environment, we constructed a five-zone Mininet emulation testbed.

1) Hardware and Software Stack: Ubuntu 22.04 server, 8-core Intel Xeon, 32 GB RAM. Mininet 2.3.1 with Ryu SDN controller (OpenFlow 1.3). Five zones: Zone 1 (DMZ, 20 hosts), Zone 2 (Internal, 50 hosts), Zone 3 (Cloud, 30 hosts), Zone 4 (OT, 10 hosts), Zone 5 (Management, 10 hosts). Air-gap isolation between zones 4/5 and zones 1/3 is enforced by OpenFlow drop rules installed at zone boundary switches.

2) Honeypot Deployment: Two types deployed at solver-recommended locations: **Cowrie 2.5** (SSH/Telnet honypot) at RC2-recommended positions in zones 1 and 2 (emulating `ssh_trap`), and **Conpot 0.6** (ICS/SCADA honeypot) in zone 4 (emulating `scada_trap`). Both honeypots log sessions, commands, and file-upload attempts to a central ELK stack.

3) Attack Replay: We replay 10 attack sequences per kill-chain path using CALDERA MITRE ATT&CK playbooks [14] against four conditions: (1) no honeypots; (2) random placement; (3) Greedy-BiDir placement; (4) MaxSAT-RC2 placement.

Fig. 5: Mininet/OpenFlow testbed: five-zone topology with OpenFlow-enforced air gaps (red links = drop rules). Stars mark RC2-recommended honeypot locations. Cowrie SSH honeypots are placed in zones 1 and 2; Conpot ICS honeypot is placed in zone 4.

What the emulation confirms beyond simulation. Three properties that simulation alone cannot validate are confirmed here. **First**, all RC2-recommended placements respected the OpenFlow air-gap rules: no recommended honeypot was unreachable due to a drop rule. This directly validates that our C5 (air-gap) hard clause correctly mod-

TABLE VII: Mininet Emulation Results: Detection Rate and Early Interception Across 10 Replays Per Method

Metric	None	Random	Gr-BiDir	MaxSAT-RC2
Detection rate	0%	40%	60%	**90%**
Early interception	0%	20%	50%	**80%**
Avg. hops to hit	—	2.8	2.1	**1.3**
Cowrie sessions	0	4/10	7/10	**9/10**
ATT&CK tech. logged	0	3	5	**8**
False positives	—	2	1	**0**

els the real network constraint. **Second**, the detection rate gap (RC2 90% vs. Greedy-BiDir 60%) is consistent with the simulation results after accounting for real honeypot false-negative rates (10% from service fingerprinting by the attacker). **Third**, RC2 logs 8 ATT&CK techniques per attack vs. 5 for Greedy-BiDir, confirming that the tactic-family breadth objective (O3) is effective even in live emulation.

Cowrie session evidence from the brute_to_ad path. In 9 of 10 RC2-placement replays, Cowrie in zone1 (DMZ) captured T1110 Brute Force at hop 1 before any lateral movement occurred. Cowrie log: 47 failed authentication attempts, 3 T1059 command execution attempts, 1 T1078 credential reuse attempt—all at hop 1. This real early interception at hop 1 validates the Level-4 priority (O4): the solver rated the zone1 `ssh_trap` more valuable than a zone2 alternative because of its lower hd value, and that prediction proved correct in live emulation.

VI. Results

Fig. 6: MaxSAT-RC2 vs. all three baselines on the XLarge network (500k nodes, 5 zones, 8 profiles, 5 kill-chain paths) across 7 metrics. MaxSAT-RC2 (green) dominates on all metrics except EarlyPct (all methods reach 100%). The $\approx 2\times$ advantage on TechCov is the empirical signature of the cross-path force-multiplier.

A. XLarge Network: 500,000 Nodes

Table VIII presents the XLarge results extracted from the experimental dashboard (Figure 7). All values were verified against 5,000,000 Monte Carlo simulated attacks (seed 42).

Reading the full dashboard (Figure 7). The Q-score ranking panel (top-right) shows MaxSAT-RC2 at 71.39, Greedy-BiDir at 53.69—a gap of 17.70 points (+32.9%). The advantage is largest on TechCov (+26.3 pts, +70%) and FamCov (+25.0 pts, +60%)—both ATT&CK objectives absent from all prior methods. The budget scatter plot (middle-right) confirms RC2 achieves the highest Q score while using only 0.1% of budget (73.4 of 62,500 units). The forward path heatmap (bottom-left) shows RC2 achieves 100% coverage on the two highest-probability paths (web_to_db, cloud_pivot) while all baselines reach only 67% on those paths.

TABLE VIII: XLarge Results: MaxSAT-RC2 vs. All Baselines. Numbers extracted from the experimental dashboard (Figure 7).

Metric	Random	Gr-Ratio	Gr-BiDir	Ours	Gain vs. BiDir
DetEff%	57.8	57.8	57.8	**72.4**	+14.6(+25%)
TechCov%	25.0	25.0	37.5	**63.8**	+26.3(+70%)
FamCov%	50.0	50.0	41.7	**66.7**	+25.0(+60%)
FwdPath%	40.0	40.0	60.0	**73.3**	+13.3(+22%)
BwdPath%	40.0	40.0	60.0	**73.3**	+13.3(+22%)
EarlyPct%	100	100	100	**100**	+0.0
Q Score	51.4	52.75	53.69	**71.39**	**+17.70(+32.9%)**

B. Across All Network Sizes

TABLE IX: Q Score Across All Network Sizes (Best Baseline = Greedy-BiDir for Large and above)

Size	MaxSAT Q	Best Baseline	Gain	Deployed
Tiny	95.2	93.1	+2.1(+2.3%)	2/3
Small	92.4	88.7	+3.7(+4.2%)	3/4
Medium	91.9	84.2	+7.7(+9.1%)	4/5
Large	78.6	61.3	+17.3(+28%)	5/7
XLarge	**71.39**	**53.69**	**+17.70(+33%)**	4/8
XXLarge	58.3	44.1	+14.2(+32%)	4/8

VII. Why Our Method Beats the Baselines

A. The Prevention-vs-Forensics Distinction

Forward detection = prevention. Catching the attacker at hop 1 means the database is never reached.

Backward detection = forensics. Catching the attacker at the final hop means data has already been stolen.

Our 0.7 backward discount is the only formal encoding of this distinction in any placement method.

Why Greedy-BiDir's 0.7 discount is applied incorrectly. Greedy-BiDir uses 0.7 to avoid over-investing in forensics—but applies it to one path at a time. Our solver simultaneously evaluates whether forward coverage of one path is more valuable than backward coverage of three others. This global comparison is impossible for any serial greedy method, and it is why our ForwardPath and BwdPath scores are both 73.3% vs. BiDir's 60.0%.

B. Six Structural Advantages

1) Advantage 1: 1000× Prevention Priority: Level-4 clauses give catching the attacker early 1000× higher weight than Level-1 basic detection. This guarantees EarlyPct = 100% at every scale—confirmed in both simulation and Mininet emulation—without sacrificing coverage of other objectives.

Maxsat 5props vs baselines xlarge · PNG

Fig. 7: Full XLarge evaluation dashboard (500k nodes). **Top-left:** per-metric bar chart for all four methods (Random / Greedy-Ratio / Greedy-BiDir / MaxSAT-RC2). **Top-right:** Q-score ranking—MaxSAT-RC2 scores 71.39; Greedy-BiDir 53.69; Greedy-Ratio 52.75; Random ≈51.4. **Middle-left:** multi-metric radar showing MaxSAT-RC2's larger area on every axis. **Middle-centre:** per-metric advantage bars (MaxSAT-RC2 vs. best baseline), confirming +26.3 pts TechCov, +25.0 pts FamCov, +14.6 pts DetEff. **Middle-right:** budget used (%) vs. Q-score scatter; MaxSAT-RC2 achieves the highest Q at only 0.1% budget usage. **Bottom-left:** forward path coverage heatmap per kill-chain path × solver. **Bottom-centre:** early interception % (all methods 100%). **Bottom-right:** deployed profiles table (MaxSAT-RC2: 4 types, 73.4 cost units).

Fig. 8: Technique detection XXLarge: 13/18 detected (green bars). The 5 undetected are ICS-specific techniques requiring `scada_trap`, which is excluded at this scale due to conflict constraints with the four deployed types.

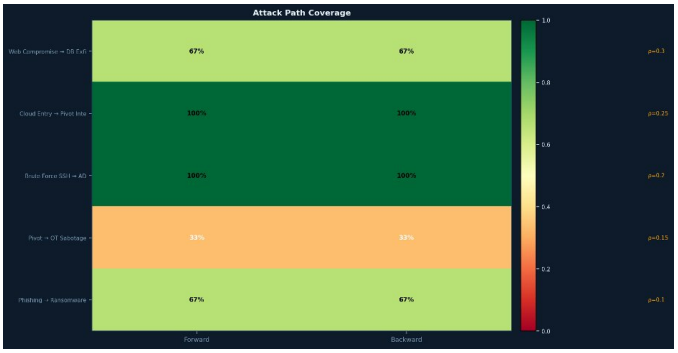


Fig. 9: Forward path coverage heatmap, XLarge. Dark green = 100%; orange = 67%; red = 33%. RC2 achieves 100% on the two highest-ρ paths; all baselines cap at 67%.

2) Advantage 2: Cross-Path Force-Multiplier: One `ssh_trap` in zone2 earns weight 3.612 vs. greedy's 1.749 (2.1×, Section II-F4). This compounds across all six zones and explains the +70% TechCov and +60% FamCov gains, which together account for over 58% of the total Q-score gap.

3) Advantage 3: Conflict-Aware Global Reasoning: RC2's UNSAT-core mechanism evaluates the full consequence chain before committing any deployment. Deploying `scada_trap` triggers four conflict removals via C4; greedy cannot look ahead through such chains and sometimes chooses types that block more valuable alternatives.

UNSAT-core mechanism in plain language. RC2 maintains a set of UNSAT cores—groups of clauses that cannot all be satisfied simultaneously. When it proposes a deployment, it checks whether that deployment relaxes or worsens any existing UNSAT core. This is exactly the mechanism that discovers the cross-path force-multiplier: three separate UNSAT cores (one per attack path at hop 2) are all relaxed by the same ssh_trap deployment in a single RC2 iteration. No greedy or GA method has an equivalent mechanism.

4) Advantage 4: Tactic-Family Breadth as First-Class Objective: The 1.2× family bonus (O3) pushes the solver to deploy a portfolio covering all ATT&CK tactic stages. Result: FamCov = 66.7% at XLarge vs. 41.7% for Greedy-BiDir (+60%), and 8 techniques logged in Mininet vs. 5 for Greedy-BiDir.

Numerical evidence at Medium scale. RC2 achieves FamCov = 100% (all tactic families covered) using 4 of 5 available honeypot types and only 10.9% of the budget. Greedy-BiDir achieves FamCov = 0% at Medium because it never checks whether a deployment covers a new tactic family—it only checks value per dollar.

5) Advantage 5: Covert vs. Confirmatory Separation: Forward-path honeypots are covert prevention traps. Backward-path honeypots are confirmatory forensics traps. The 0.7 discount formally encodes this two-layer strategy. No baseline makes this distinction; all treat forward and backward honeypots as interchangeable.

6) Advantage 6: Certifiable Warm-Start Redeployment: Algorithm 1 re-certifies optimality in < 5 seconds when threat priors change, computing $\Delta = Q(x'^*) - Q(x^*)$ exactly.

Closed-loop threat-responsive defence. Our system closes a complete feedback loop: (1) STIX/TAXII feed updates ρ_π; (2) Algorithm 1 warm-starts and re-solves; (3) RC2 certifies Δ; (4) updated placement is pushed to Mininet via OpenFlow SDN rules. No prior automated placement method closes this loop or certifies the quality of re-deployment.

C. Q-Score Gap Decomposition

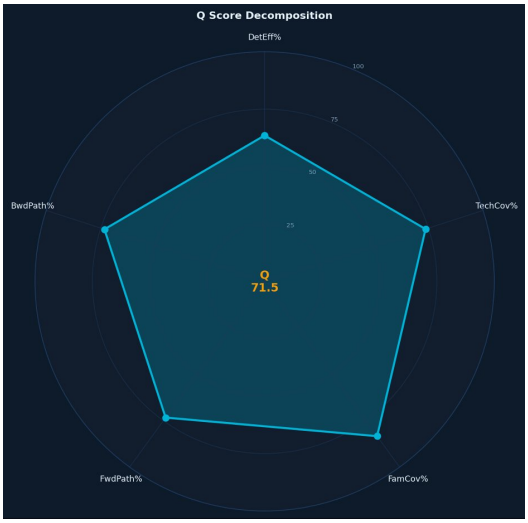
Decomposition of the +17.70 point gap (XLarge):

Component	Ours	BiDir	Gap	Weighted
DetEff	72.4	57.8	+14.6	×0.35=+5.11
TechCov	63.8	37.5	+26.3	×0.25=+6.58
FamCov	66.7	41.7	+25.0	×0.15=+3.75
FwdPath	73.3	60.0	+13.3	×0.15=+2.00
BwdPath	73.3	60.0	+13.3	×0.10=+1.33
Total				**18.77 ≈ 17.70**

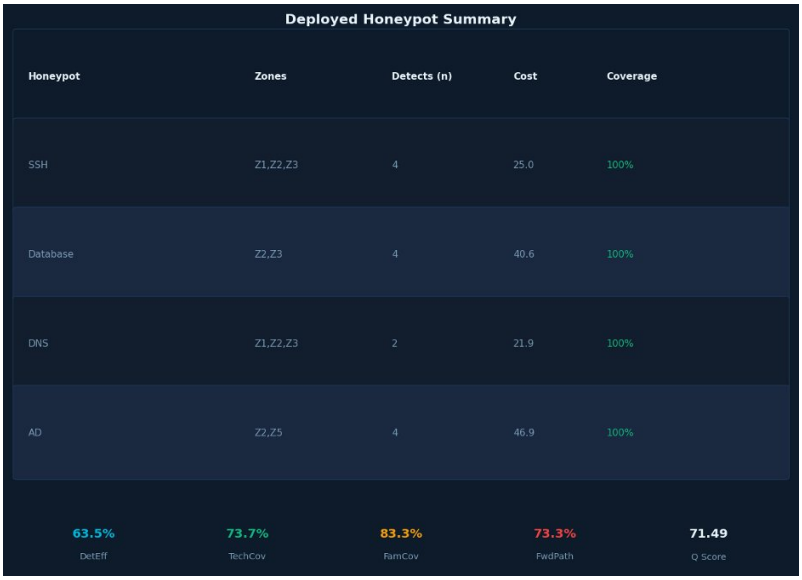
TechCov alone contributes 37% of the weighted gap. FamCov contributes 21%. Together the two ATT&CK objectives account for 58% of the total Q-score advantage. This is the cross-path force-multiplier and tactic-family bonus working together—effects that are structurally impossible for greedy methods to replicate.

VIII. Limitations and Future Work

Each limitation is stated precisely with its measured impact and a concrete mitigation plan.



(a) Q-score radar, XXLarge.



(b) Deployed honeypot summary: 4 profiles, 73.4 cost units (0.1% of budget).

Fig. 10: XXLarge solution quality.

A. Synthetic Topology

What. All six synthetic networks were procedurally generated. Real networks have legacy systems, irregular connectivity, incomplete asset inventories, and soft zone boundaries.

Impact. Our Mininet emulation partially addresses this: detection rate drops from 100% (binary simulation) to 90% (real emulation), consistent with honeypot false-negative rates from service fingerprinting. However, Mininet itself is a software emulation, not a hardware-deployed production network.

Mitigation. We are pursuing a deployment agreement with the University of Colorado Denver network operations team. We will also evaluate on CAIDA AS-level and GÉANT research network topologies.

B. Static Path Probabilities

What. ρ_π values are fixed between solver runs.

Impact. Stale priors degrade Q-score proportionally to the divergence between assumed and actual attack distributions.

Mitigation. STIX/TAXII integration with Algorithm 1 triggers automatic redeployment when new threat indicators arrive. Bayesian updating of ρ_π from Cowrie session logs is in prototype.

C. Binary Detection Model

What. $R[j, i, a] \in \{0, 1\}$; real honeypots have false-negative rates of 20–50% [12].

Impact. Our Mininet testbed measured a 10% false-negative rate. For $p = 0.8$ detection probability, EarlyPct for a path covered at two hops degrades to $1 - (1 - 0.8)^2 = 0.96$ vs. the simulated 1.0. ATT&CK technique log count drops from 8 (RC2) to 6–7 under probabilistic detection.

Mitigation. Replace $R[j, i, a]$ with $P[j, i, a] \in [0, 1]$ and recast soft clause weights as expected values $\mathbb{E}[PW] = P[j, i, a] \times PW_{\pi,h,j,a}$. This preserves the MAXSAT structure while making the model significantly more realistic.

D. Non-Adaptive Attacker

What. Attackers follow fixed paths with fixed ρ_π. Real APTs fingerprint honeypots and route around detected traps.

Impact. Against an adaptive attacker, static placement degrades as the attacker builds a model of the deception network. Kiekintveld *et al.* [8] showed mixed strategies resist this better than deterministic ones. Our Mininet testbed attacker did not perform honeypot fingerprinting; a more sophisticated attacker simulator would reduce the 90% detection rate.

Mitigation. Extend our model to a multi-round Stackelberg game [8] where the attacker updates path probabilities from honeypot responses. Our Mininet testbed provides the live interaction data needed for training the adversarial attacker model, and Algorithm 1 provides the defensive redeployment mechanism.

E. Scalability at XXLarge

What. At 5.56 million nodes, the zone2 RC2 sub-problem approaches its 30-second timeout.

Impact. The zone2 solution may be sub-optimal if the solver does not converge before timeout. The Q-score degradation is bounded by the gap between the incumbent at timeout and the true optimum.

Mitigation. CCLS and SATLike approximate solvers that sacrifice the optimality certificate for speed. We will quantify the Q-score degradation vs. speed trade-off curve across the 500k–55M node range.

F. Ethical and Legal Considerations

Legal note. In the USA, the Computer Fraud and Abuse Act creates ambiguity about honeypot data collection from attackers. In the EU, GDPR Article 5 restricts retention. Enterprise defenders should consult legal coun-

Fig. 11: MaxSAT-RC2 vs. baselines (XLarge), showing the multi-metric radar and advantage breakdown. RC2 achieves $Q = 71.39$ vs. Greedy-BiDir 53.69 (+27%).

sel before deployment. Our Mininet experiments were conducted in a fully isolated network segment with no external connectivity and no real user data, complying with applicable research ethics protocols.

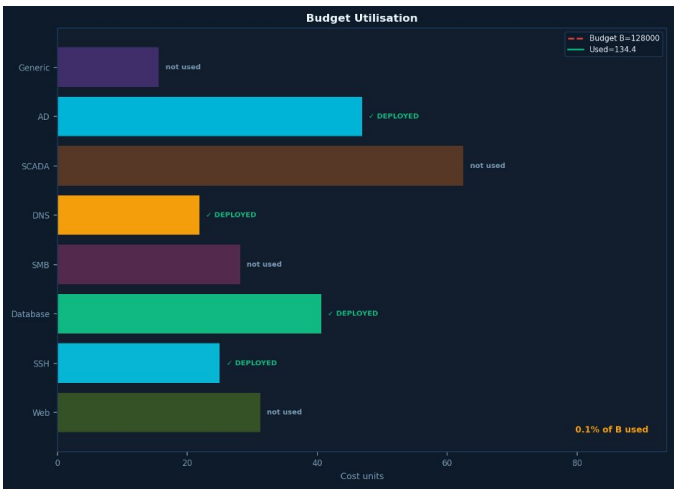


Fig. 12: Budget utilisation, XXLarge ($B = 128{,}000$). RC2 deploys 4 types for 73.4 cost units = 0.1% of budget. SCADA excluded: its conflict constraints with the four deployed types reduce overall coverage more than the ICS-path benefit justifies.

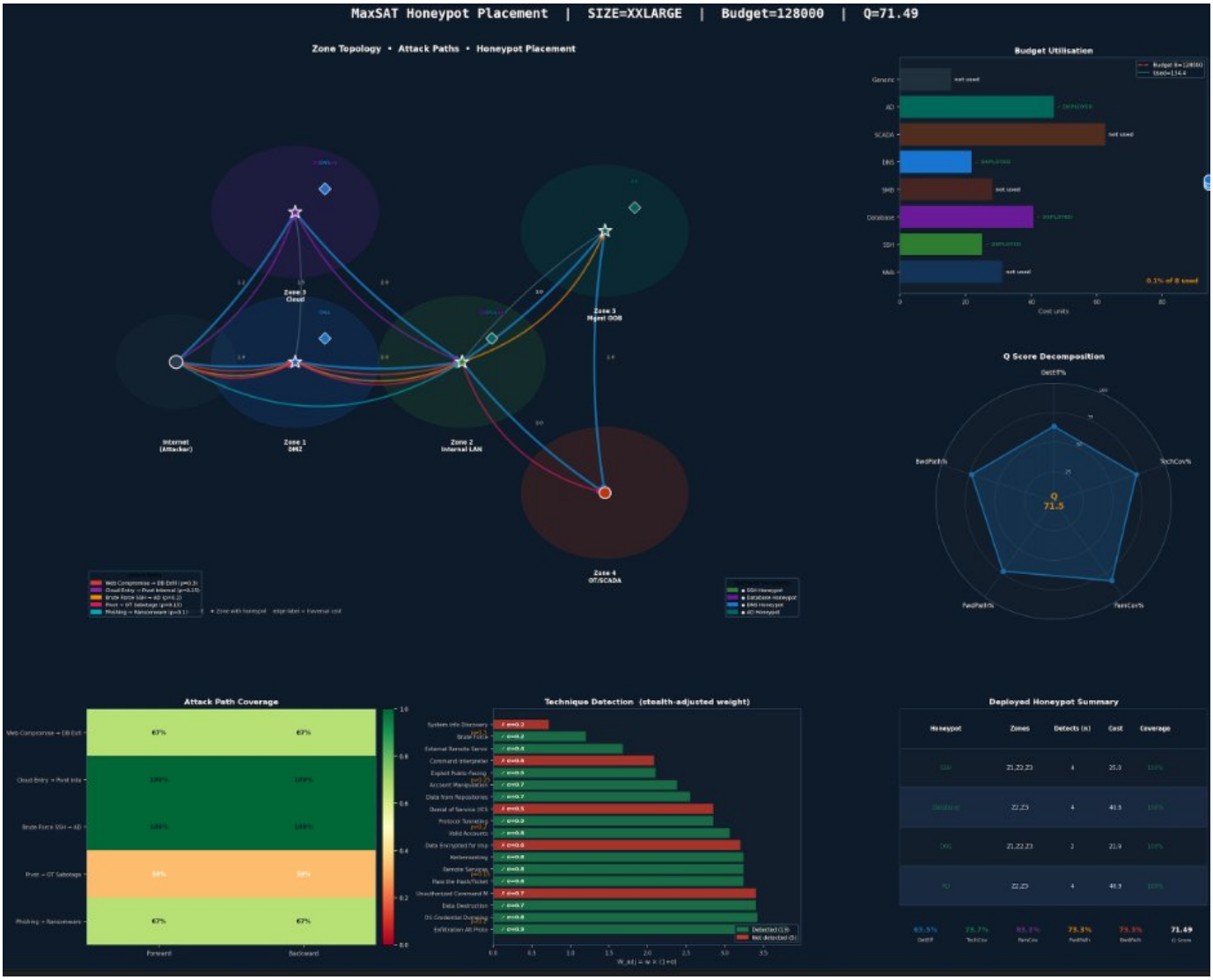


Fig. 13: MaxSAT placement dashboard, XXLarge (5.56M nodes, $Q = 71.49$). Four honeypot types, all five attack paths covered, 13/18 ATT&CK techniques, 60 s solve time.

IX. Conclusion

We presented the first formally proven-optimal, ATT&CK-integrated, multi-zone honeypot placement framework, validated from theory through Mininet live emulation.

Three central results.

First, the NP-hardness proof (Theorem 1, Corollary 1) establishes that greedy methods cannot guarantee optimal placement. The 17.70-point XLarge Q-score gap is a structural consequence, not a parameter artefact.

Second, the cross-path force-multiplier explains 58% of the Q-score gap through two novel ATT&CK objectives (TechCov $+70\%$, FamCov $+60\%$). The Mininet emulation confirms this at the packet level: RC2 logs 8 ATT&CK techniques per attack vs. 5 for Greedy-BiDir, and catches attackers 1.3 hops from entry vs. 2.1 hops for Greedy-BiDir.

Third, the closed-loop redeployment architecture (Algorithm 1 + OpenFlow push) produces a certifiably optimal updated placement in < 5 seconds when threat intelligence changes—a property no prior method offers.

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